different sets of basis vectors. For example, in the case of a single spin, the inner products

$$\psi(u) = \langle u|\Psi\rangle$$

and

$$\psi(d) = \langle d|\Psi\rangle$$

define the wave function in the σ_z basis, while

$$\psi(r) = \langle r|\Psi\rangle$$

and

$$\psi(l) = \langle l|\Psi\rangle$$

define the wave function in the σ_x basis.

An important feature of the wave function follows from the fact that the total probability sums to one:

$$\sum_{a,b,c,\ldots} \psi^*(a,b,c,\ldots)\ \psi(a,b,c,\ldots) = 1.$$

5.1.3 A Note About Terminology

The term *wave function*, as used in this book, refers to the collection of coefficients (also called components) that multiply the basis vectors in an eigenfunction expansion. For example, if we expand a state-vector |Ψ> as follows,